



AI-Governed Enterprise Knowledge Assistant

AI Ideathon 2026 Submission Document

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Live Demo: <https://aiideathon.vercel.app>

Source Code: https://github.com/meetpstewart/AI_Ideathon

Backend API: <https://ai-knowledge-backend-655994006172.us-central1.run.app>

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Live Deployment Summary

Component	URL / Location
Web Application	https://aiideathon.vercel.app
Backend API	https://ai-knowledge-backend-655994006172.us-central1.run.app
Slack Bot	/ask command — AI_Documentation_Assistant workspace
Source Code	https://github.com/meetpstewart/AI_Ideathon
Frontend Hosting	Vercel (auto-deploys on push to main)
Backend Hosting	Google Cloud Run (us-central1)

1. Problem Statement

Enterprise documentation at Stewart is spread across hundreds of PDF and Word files — difficult to search, dependent on subject matter experts, and inaccessible to most employees. When teams attempt to use general-purpose AI tools (like ChatGPT) to answer internal questions, they face two critical risks:

- Hallucination — the AI answers from its training data, not from Stewart's actual documents, producing plausible but potentially wrong information
- No traceability — there is no way to verify or audit where an AI answer came from, violating enterprise governance and compliance standards

The core problem: How do we give employees fast, accurate answers from internal documentation — without hallucination risk and with full traceability?

2. Solution Overview

The AI-Governed Enterprise Knowledge Assistant is a RAG-based (Retrieval-Augmented Generation) AI system that answers employee questions strictly from approved Stewart documentation — with every answer cited, traceable, and confidence-rated.

How It Works

Instead of asking an AI to guess from its training data, the system first searches approved documents for the most relevant passages, then hands those passages to the AI and instructs it: "Answer using only this." The AI responds with the answer and a citation pointing back to the exact document it used.

Comparison — Our System vs Generic AI

Feature	Generic AI (ChatGPT, etc.)	Our System
Source of answers	AI training data (internet)	Approved Stewart documents only
Hallucination risk	High	Eliminated by design
Traceability	None	Every answer cited to source document
Confidence rating	None	FULL / PARTIAL / NO SUPPORT
Enterprise governance	None	Built-in — no data leaves GCP
Conversation memory	Session-based	Multi-turn with context
Access channels	Web only	Web UI + Slack bot

Key Features Delivered

- Cited answers — every response includes numbered citations [1] linked to source documents
- Confidence scoring — answers rated FULL SUPPORT, PARTIAL SUPPORT, or NO SUPPORT
- Multi-turn conversation — follow-up questions maintain full context (last 3 turns)
- Suggested follow-ups — AI proactively suggests 3 related questions after each answer
- Thumbs up / down feedback — per-answer helpfulness rating
- Session analytics — real-time dashboard: confidence breakdown, satisfaction rate, top referenced documents
- Slack bot integration — /ask slash command works in any Slack channel
- Stewart-branded UI — full enterprise design with official branding, colours, and logo

3. Architecture Design

Technology Stack

Layer	Technology	Purpose
Frontend	React 19 + Vite, hosted on Vercel	Web UI — chat, analytics, branding
Backend	Python FastAPI on Google Cloud Run	REST API, RAG orchestration
AI Search	Google Discovery Engine (Vertex AI)	Semantic document search + retrieval
AI Model	Gemini 2.5 Flash (Vertex AI)	Answer synthesis + follow-up generation
Embeddings	text-embedding-004 (Vertex AI)	Document vectorisation (offline)
Storage	Google Cloud Storage	Approved documents + embeddings
Messaging	Slack API (slash commands)	Bot integration for /ask

Google Cloud Services Used

GCP Service	Role in the System
Vertex AI — Gemini 2.5 Flash	Generates grounded answers from retrieved document context
Vertex AI — Discovery Engine	Indexes documents and performs semantic search at query time
Vertex AI — text-embedding-004	Converts document chunks to vectors for semantic indexing
Cloud Run	Hosts FastAPI backend as a serverless, auto-scaling container
Cloud Storage (GCS)	Stores approved documents and generated embeddings
Cloud Build	Builds Docker container images on every deployment
Artifact Registry	Stores built container images for Cloud Run
Cloud IAM	Controls service-to-service authentication and permissions

RAG Pipeline — How Answers Are Generated

```

User Question
  |
  v
Discovery Engine searches indexed documents
  |
  v
Top 3 most relevant document passages retrieved
  |
  v
Numbered context block built:
[1] Source: Cloud Infrastructure — "The Quarter Rack has 1440 GB RAM..."
[2] Source: App Service Guide   — "...
  |
  v

```

```

Gemini prompt: "Answer ONLY using the numbered sources. Use citation [1]."
```

|
v

```

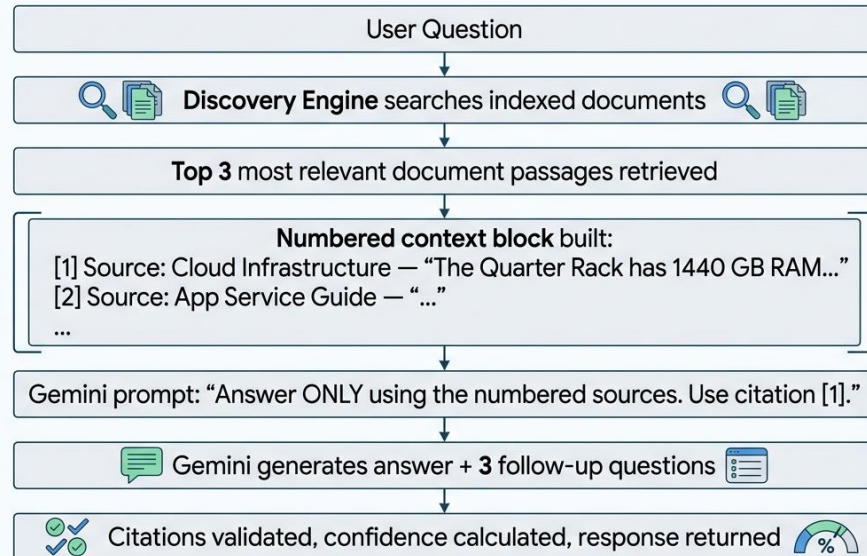
Gemini generates answer + 3 follow-up questions
```

|
v

```

Citations validated, confidence calculated, response returned
```

RAG Pipeline — How Answers Are Generated

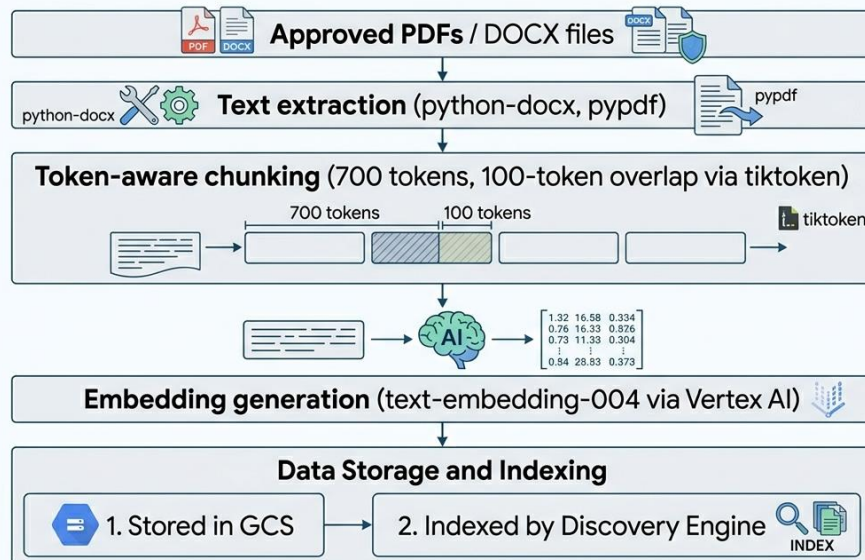


Document Ingestion Pipeline (Offline)

```

Approved PDFs / DOCX files
-> Text extraction (python-docx, pypdf)
-> Token-aware chunking (700 tokens, 100-token overlap via tiktoken)
-> Embedding generation (text-embedding-004 via Vertex AI)
-> Stored in GCS -> Indexed by Discovery Engine
```

Document Ingestion Pipeline (Offline)



Confidence Scoring Logic

Confidence Level	Condition	Meaning
FULL SUPPORT	2 or more documents cited	Answer corroborated by multiple independent sources
PARTIAL SUPPORT	Exactly 1 document cited	Answer sourced from a single document
NO SUPPORT	No documents cited	Could not be grounded — fallback message shown

4. Working Demonstration

Live URLs

	URL
Web Application	https://aiideathon.vercel.app
Backend Health	https://ai-knowledge-backend-655994006172.us-central1.run.app
Slack	/ask in AI_Documentation_Assistant workspace

Demo

Step 1 — Open the web app

Navigate to <https://aiideathon.vercel.app> on any browser or mobile device.

Step 2 — Ask a question

Click a starter chip or type: "What are the RMI ports for distributed installation?"

Observe: Answer with [1] citation superscripts

Observe: FULL / PARTIAL SUPPORT confidence badge

Observe: Expandable source document card

Observe: 3 follow-up question chips

Step 3 — Click a follow-up chip

Click one of the suggested follow-up questions.

Observe: Answer references the previous turn (multi-turn memory active)

Step 4 — Rate the answer

Click the thumbs up or thumbs down button on any response.

Step 5 — View Analytics tab

Click Analytics in the left sidebar.

Observe: Questions asked, answers generated, satisfaction rate, confidence breakdown, top documents

Step 6 — Slack demo

In the Slack workspace, type in any channel:

/ask What is the total RAM capacity for Exadata System Configuration?

Observe: Instant acknowledgment < 1 second

Observe: Full answer posted with confidence emoji, sources, follow-up suggestions

Governance Demo

- Ask a question NOT covered in the documents → System says "The information is not available in the approved documentation" — it does not hallucinate
- Show the NO SUPPORT confidence badge when triggered
- Show source cards — every claim traceable to an exact document with title and link
- Show Analytics tab — confidence breakdown proves governance is measurable

5. Competitive Differentiation - Why Not Just Use Microsoft 365 Copilot?

A common question when evaluating this project: "Doesn't Microsoft 365 Copilot already do this?"

The short answer: Copilot solves a similar problem space, but in a fundamentally different — and less governed — way.

Feature Comparison

Feature	M365 Copilot	Our System
Cost	~\$30/user/month (\$360/user/year)	GCP pay-per-use (cents per query)
Document scope	Searches ALL files user can access	Only explicitly approved documents
Citation level	General "based on your files"	Numbered [1] citations per sentence
Confidence scoring	None	FULL / PARTIAL / NO SUPPORT
Governance control	Microsoft controls the stack	100% isolated in Stewart's GCP project
Approval workflow	No document gating	Only approved docs can be cited
Custom branding	Microsoft UI only	Full Stewart brand identity
Session analytics	None	Confidence breakdown + satisfaction rate
Follow-up suggestions	Basic	Context-aware, topic-specific chips
Vendor dependency	Deep Microsoft lock-in	Portable — any GCP service swappable
M365 license required	Yes — E3/E5 + Copilot add-on	No — open web + Slack
Data residency	Microsoft processes your data	Data never leaves your GCP project

The Core Governance Difference

Microsoft 365 Copilot searches everything a user has permission to access — SharePoint, OneDrive, emails, Teams chats. There is no document approval layer. A user asking about cloud infrastructure might get an answer pulled from an outdated email thread, an unapproved draft, or an irrelevant HR document.

Our system inverts this: only documents that have been explicitly approved, ingested, and indexed can ever appear in an answer. This is a fundamentally different trust model — one that aligns with enterprise compliance, legal defensibility, and audit requirements.

The Citation Difference

Copilot tells you an answer came "from your SharePoint files."

Our system tells you: "The Quarter Rack has 1440 GB RAM [1] — sourced from Cloud Infrastructure Guide, page 4." Every factual claim is numbered, every number links to a specific document card with title, URI, and the exact passage used. That is citation-level accountability that Copilot does not offer.

The Cost Difference

Scenario	M365 Copilot	Our System
100 users, 1 year	~\$36,000	~\$50–200 (GCP usage only)
Requires M365 E3/E5 license	Yes (~\$36/user/month extra)	No
Per-query cost model	Flat subscription	Pay only for what you use

Summary

This project demonstrates a production-deployed, enterprise-grade AI system built entirely on Google Cloud Platform during the AI Ideathon 2026. It directly addresses a real business problem at Stewart — knowledge accessibility and AI governance — using industry-standard patterns (RAG, grounding, citation tracking) deployed on GCP-native infrastructure.

The system is live, accessible from any device, and fully functional at <https://aiideathon.vercel.app> and via the Slack bot — not a prototype or mockup.